

Margherita Calderan, Ph.D.

My research interests include computational models of cognition, Bayesian statistics, and metascience.

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Current Position

Research Assistant

Department of Developmental Psychology and Socialization, University of Padova, Italy

May 2026 - Present

- Number-space association: A comparative, developmental and neurobiological approach
- PI: Prof. Silvia Benavides

Research

Postdoctoral Researcher

Department of General Psychology, University of Padova, Italy

Oct. 2024 - Mar. 2026

- Tails or types? Testing the dimensional hypothesis in neurodevelopmental disorders
- Supervisor: Prof. Enrico Toffalini

Education

Ph.D. in Psychological Sciences - cum laude

University of Padova, Italy

Defense: 06/03/2025

- Thesis: Balancing stability and flexibility in cognitive control and learning: the effects of arousal
- <https://researchdata.cab.unipd.it/1437/>
- Supervisors: Prof. Roberta Sellaro, Prof. Nicola Cellini

Postgraduate Internship

Psychophysiology Lab, University of Padova, Italy

Jan. 2021 - Dec. 2021

- Investigated the role of sleep characteristics in false memory formation and emotional reactivity.

- Supervisors: Prof. Andrea Spoto, Prof. Nicola Cellini

M.Sc. in Cognitive Neuroscience and Clinical Neuropsychology - 110/110 cum laude

University of Padova, Italy

Oct. 2018 - Dec. 2020

- Thesis: Sleep characteristics in adults with high functioning autism: a polysomnographic study
- Supervisor: Prof. Nicola Cellini

Curricular Internship

Psychophysiology Lab, University of Padova, Italy

Dec. 2019 - Aug. 2020

- Analyzed polysomnographic data to examine sleep architecture in young adults with high-functioning autism.
- Collaborated with the University of Trento and the University of Tübingen.
- Supervisor: Prof. Nicola Cellini

B.Sc. in Cognitive Psychology and Psychobiology - 109/110

University of Padova, Italy

Sept. 2015 - July 2018

- Thesis: Gender differences in the long-term consolidation of emotional memories
- Supervisor: Prof. Michela Sarlo

Abroad

Visiting Postdoctoral Researcher

Department of Biomedical Data Sciences, Leiden University Medical Center, the Netherlands

1.5 months (Sept. 2025 - Oct. 2025)

- Research on Bayesian approaches to the problem of multiplicity.
- Supervisor: Prof. Jelle Goeman

Visiting Ph.D. Student

Cognitive Psychology Unit, University of Leiden, the Netherlands

4 months (Feb. 2024 - Jun. 2024)

- Worked on a research project investigating how mind-wandering affects cognitive flexibility, applying computational models (<https://psyarxiv.com/quhns>) to analyze behavioral data.
- Supervisor: Prof. Bryant Jongkees

Research Intern

Donders Sleep and Memory Lab, Donders Institute, the Netherlands

2 months (Jun. 2019 – Aug. 2019)

- Assisted with a project on the association between affective memory bias, medial prefrontal schema processing, and sleep in depressed patients and healthy controls.
- Assisted with a project on lucid dream induction.
- Supervisor: Prof. Martin Dresler

Erasmus +

Jaume I University, Spain

5 months (Feb. 2015 – July 2015)

- Completed a semester-long academic exchange as part of my undergraduate studies.

Publications

Published

1. Mazzoni, N., Kurz, E.V., Cellini, N., Ciringione, L., **Calderan, M.**, Gallitto, G., Falco, S., Venuti, P., Zinke, K., Born, J., & Caria, A. (2026). Aperiodic and periodic neural activity during sleep in autism spectrum disorders. *BMC Medicine*. <https://doi.org/10.1186/s12916-026-04771-6>
2. **Calderan, M.**, Visalli, A., Degasperi, G., Toffoli, L., Sellaro, R., & Cellini, N. Predictions under sleep restriction: The effects of partial sleep deprivation on learning and inhibitory control. *Royal Society Open Science*. <https://doi.org/10.1098/rsos.252345>. **Preprint:** https://doi.org/10.31234/osf.io/x7m4z_v3; **OSF** repo: <https://osf.io/6fxmh/>
3. **Calderan, M.**, Jongkees, J.B., Gambarota, F., Pastore, M., Cellini, N., & Sellaro, R. (2025) Isolating working memory gating processes under stress. *Collabra::Psychology*. <https://doi.org/10.1525/collabra.140750>; **OSF** repo: <https://osf.io/d43vs/>
4. **Calderan, M.**, Toffalini, E., Mirandola, C., & Cellini, N. (2025). Sleep does not affect false memory production on a script-based test. *Neuropsychologia*. <https://doi.org/10.1016/j.neuropsychologia.2025.109210>; **OSF** repo: <https://osf.io/kwycq/>
5. Toffoli, L., **Calderan, M.**, Del Popolo Cristaldi, F., Duma, G. M., Calcagnì, A., Pastore, M., Tarantino, V., & Mento, G. (2025). Can I afford one more candy? How motivational contexts shape adaptive cognitive control in children. *Developmental Psychology*. <https://doi.org/10.1037/dev0001976>; **OSF** repo: <https://osf.io/shuq2/>
6. Sobol, M., Blachnio, A., **Calderan, M.**, Degasperi, G., Mioni, G., Hryhorchuk, I., Plucińska, E., Stasiniewicz, J., Kwapiński, J., Jankowski, K., & Cellini, N. (2025). Balanced time perspective as a factor mitigating the negative consequences of sleep disruptions in the perinatal period. *Chronobiology International*, 1–12. <https://doi.org/10.1080/07420528.2025.2490499>
7. **Calderan, M.**, & Visalli, A. (2024). Challenges of meta-learning and rational analysis in large worlds. *Behavioral and Brain Sciences*, 47, e148. <https://doi.org/10.1017/S0140525X24000128>.

Under Review

1. Visalli, A., Calistrone, F. M., **Calderan, M.**, Donnarumma, F., Zorzi, M., & Ambrosini, E. A computational framework for integrating Predictive processes with evidence Accumulation Models (PAM). Under review in *Behavior Research Methods*. **Preprint:** <https://doi.org/10.48550/arXiv.2411.13203>.
2. Toffalini, E., **Calderan, M.**, Pagan, R., Tobia, V., & Sonuga-Barke, E. Tails or Types? A Critical Systematic Review of Taxometric Studies of Neurodevelopmental Conditions. Under review in *JCPP Advances*. **Preprint:** <https://osf.io/ys5ad/files/bem36>.
3. Feraco, T., **Calderan, M.** & Pellegrino, G. (2025). The Living SEB Skills Project: A Living Resource for Review and Meta-Analysis of Social, Emotional, and Behavioral Skills. Under review in *European Journal of Personality*. **Preprint:** https://doi.org/10.31234/osf.io/ewvuf_v1.

Preregistration

1. Sirota, M., Šrol, J., Lisi, M., Juanchich, M., Guglani, K., Silverstein, P., ... **Calderan, M.**, ... Schmidt, K. (2025, January 14). Mapping and Increasing Error Correction Behaviour in a Culturally Diverse Sample. **Preprint:** <https://doi.org/10.17605/OSF.IO/WN4VX>.
2. Toffalini, E., **Calderan, M.**, Sità, L., Feraco, T., & Gambarota, F. Link Functions and Generalized Linear Models in Interaction Testing within Psychological Research. <https://osf.io/sjdu4>

In Preparation

1. **Calderan, M.**, Gambarota, F., Toffalini, E., & Altoè, G., Adjusting the Probability of Direction for Multiple Testing.
2. Goeman, J., **Calderan, M.**, & Solari, A., Bonferroni for Bayesians: Multiplicity Correction Based on Joint Credibility
3. Barbiero, J., **Calderan, M.**, Zeilinga, K., & Mioni, G., The contribution of the cerebellum in time perception: A systematic review and meta-analyses.
4. Toffoli, L., **Calderan, M.**, Stefanelli, G., Duma, G.M., Del Popolo Cristaldi, F., Duma, G., Pastore, M., Frisina, S., Tarantino, V., & Mento, G., I am talking to you! Engagement modulates the effects of digital content exposure on preschoolers' adaptive cognitive control.
5. Bastianello, T., **Calderan, M.**, Loschiavo, A.G.M., Nascimben, C., Visibelli, E., Reoyo-Serrano, N., Altoè, G., & Benavides-Varela, S. Domestic Media Exposure Delays Attainment of Babbling in Infants While Adult Input Accelerates Progress.
6. **Calderan, M.**, van Dooren, R., Sellaro, R., & Jongkees, J.B., Dynamic and contextual modulation of constraints on cognitive control.

Published Abstract

1. **Calderan, M.**, Visalli, A., Sellaro, R., & Cellini, N. (2024). Restless Sleep, Uncertain Minds: Learning and Inhibitory Control Under Partial Sleep Deprivation. Proceedings of the Annual Meeting of the Cognitive Science Society, 46. <https://escholarship.org/uc/item/65b2f0n7>.
2. **Calderan, M.**, Visalli, A., Sellaro, R., & Cellini, N. (2024). The Effects of Partial Sleep Deprivation on Adaptive Cognitive Control. *Sleep Medicine*, <https://doi.org/10.1016/j.sleep.2023.11.156>.
3. **Calderan, M.**, Visalli, A., Sellaro, R., & Cellini, N. (2024). Expecting the unexpected: a closer look at the effect of partial sleep deprivation on learning and inhibitory control. *Journal of Sleep Research*, <https://doi.org/10.1111/jsr.14291>.
4. Baldassarri, A., Colombo, C., Bergamo, D., **Calderan, M.**, Varanini, A., Ferrentino, A., D'Atri, A., & Cellini, N. (2024). Exploring the impact of oscillatory events during a nap on emotional memory consolidation. *Journal of Sleep Research*, <https://doi.org/10.1111/jsr.14291>

Conferences

1. Bastianello, T., Loschiavo, A., Nascimben, C., Reoyo-Serrano, N., **Calderan, M.**, Altoè, G., & Benavides-Varela, S. (accepted). The effects of TV and electronic devices on language: a longitudinal study. ICIS 2026 (Panama), July 2026.
2. Barbolan, S., Slaby, R., **Calderan, M.**, Knedlíková Wanková, V., & Bertamini, M. Emotional Responses to Renaissance Art: Differences Between Venetian and Bohemian Paintings. Poster accepted at the AIP - Experimental Section XXXII Conference, Venice (Italy), September 2026.
3. **Toward Robust Replicability: Modern Perspectives and Methods.** Organizer: **Calderan, M.** Speakers: Alongi, E., Garofalo, S., **Calderan, M.**, Andreella, A. Moderator: Epifania, M.O. Symposium accepted for the AIP - Experimental Section XXXII Conference, Venice (Italy), September 2026.
4. Barbolan, S., Slaby, R., **Calderan, M.**, Knedlíková Wanková, V., & Bertamini, M. Emotional Responses to Renaissance Painting: A Comparison Between Venetian and Bohemian Works. Poster accepted at the Visual Science of Art Conference (VSAC) 2026, Vision Forward: Perception, Art & Technology, London (United Kingdom), August 2026.
5. Barbolan, S., Slaby, R., **Calderan, M.**, Knedlíková Wanková, V., & Bertamini, M. Emotional and Aesthetic Experience in Venetian and Bohemian Paintings from the Renaissance. Poster presented at the International Association of Empirical Aesthetics (IAEA) 2026 Conference, Future Follows Past – Creating a Gesamtkunstwerk together for Arts and Aesthetics Research, Jena (Germany), May 2026.

6. Bastianello, T., **Calderan, M.**, Loschiavo, A., Nascimben, C., Visibelli, E., Reoyo-Serrano, N., Altoè, G., & Benavides-Varela, S. (2026). L'esposizione ai media rallenta il raggiungimento del babbling e l'input dell'adulto ne accelera il progresso. Presentazione Orale in sessione, Conferenza CLASTA, Torino (5-6 June 2026).
7. Stefanelli G., Toffoli L., **Calderan M.**, Del Popolo Cristaldi F., Duma G. M., Tarantino V., Visalli A., Sellaro R., Mento G. Apprendimento associativo implicito e volatilità ambientale nell'ADHD. **Video-Poster** at Giornate di aggiornamento sull'uso degli strumenti in Psicologia Clinica dello Sviluppo, Bologna (Italy), February 2025.
8. Stefanelli G., Toffoli L., **Calderan M.**, Duma G. M., Del Popolo Cristaldi F., Tarantino V., Visalli A., Sellaro R. & Mento G. Super Learning Bros: Developmental trajectories of probabilistic learning in volatile environment. **Talk** at the Cognitive Science Arena, Brixen (Italy), February 2025.
9. **Calderan, M.**, Visalli, A., Degasperi, G., Calistrone, F.M., Sellaro, R., & Cellini, N. Expecting the unexpected: a closer look at the effect of partial sleep deprivation on learning and inhibitory control. **Poster** presented at Sleep Europe 2024, Seville (Spain), September 2024.
10. **Calderan, M.**, Visalli, A., Degasperi, G., Calistrone, F.M., Sellaro, R., & Cellini, N. Restless Sleep, Uncertain Minds: Learning and Inhibitory Control Under Partial Sleep Deprivation. **Poster** presented at CogSci 2024: Dynamics of Cognition, Rotterdam (Netherlands), July 2024.
11. Degasperi, G., **Calderan, M.**, Mioni, G., Meisner, M., Blachnio, A., Hryhorchuk, I., Plucinska, E., Sobol, M., & Cellini, N. The role of sleep, circadian rhythms, balanced time perspective on perinatal depression symptoms. **Poster** at the Cognitive Science Arena, Brixen (Italy), February 2024.
12. Toffoli, L., **Calderan, M.**, Duma, G. M., Frisina, S., Tarantino V., & Mento, G. I'm talking to you! Interactive digital content effects on children's adaptive cognitive control. **Talk** at the Cognitive Science Arena, Brixen (Italy), February 2024.
13. **Calderan, M.**, Visalli, A., Sellaro, R., & Cellini, N. The Effects of Partial Sleep Deprivation on Adaptive Cognitive Control. **Poster** presented at World Sleep Congress, Rio de Janeiro (Brazil), October 2023.
14. **Calderan, M.**, Visalli, A., Sellaro, R., & Cellini, N. The sleep-deprived predictive brain: modeling the effect of partial sleep deprivation on adaptive cognitive control. **Talk** at the XXIX Conference of the Italian Association of Psychology (AIP) - Experimental Psychology Section. Lucca (Italy), September 2023.
15. **Calderan, M.**, Cellini, N., Sellaro, R., Stressed Minds: Unraveling Acute-Stress Effects on Working Memory Processes. **Mini-Talk** at the Cognitive Science Arena, Brixen (Italy), February 2023.
16. Toffoli, L., **Calderan, M.**, Del Popolo Cristaldi, F., Duma, G. M., Calcagnì, A., Pastore, M., Tarantino, V., & Mento, G. Can I afford one more candy? How motivational contexts shape adaptive cognitive control in children. **Talk** at the Cognitive Science Arena, Brixen (Italy), February 2023.

17. **Calderan, M.**, Cellini, N., Sellaro, R., Exploring the Influence of Stress on Working Memory Gating Mechanisms. **Mini-Talk** at the XXX Conference of the Italian Association of Psychology (AIP). Padova (Italy), September 2022.
18. **Calderan, M.**, Cellini, N., Sellaro, R., Understanding stress-induced Effects on Working Memory gating mechanisms. **Poster** presented at the 22nd meeting of the European Society for Cognitive Psychology (ESCoP), Lille (France), August 2022.

Pre-conference workshops

- **Generalized Linear Models for Psychology: Understanding, Simulating, and Planning Your Analyses.** Pre-conference Workshop **Cognitive Science Arena**, Brixen (Italy). 5 February 2026. Hands-on methodological workshop held with Filippo Gambarota, PhD. <https://psicostat.github.io/glm-workshop/>
- **A Practical Introduction to Evidence Accumulation Models in Visual Perception Research.** European Conference on Visual Perception, Mainz (Germany). 24 August 2025 (2.5 hrs.). Hands-on methodological workshop held with Carolina Maria Oletto, PhD. <https://mar-cald.github.io/eam-workshop/eam-home.html>.
- **Introduction to Reproducible Science.** Pre-School workshop *Replicability Crisis in Science*, Padova (Italy). 6 June 2025 (2 hrs.). Material: <https://github.com/Mar-Cald/repro-pre-school>.

Invited talks

- **Bayesian inference and multiple comparisons: From idealization of methods to conscious practice.** Cognitive Neuroscience Seminars, Department of General Psychology, University of Padova, Italy. 4th December 2025.
- **From Cognitive Psychology to Bayesian Multiple Testing.** Medical Statistics Meeting, Department of Biomedical Data Sciences, Leiden University Medical Center, Leiden, The Netherlands. 23rd October 2025.
- **Bayesian Multiple Testing.** STAT-Talk Series, Methodology and Statistics Department, Institute of Psychology, Leiden University, Leiden, The Netherlands. 18th September 2025.

Awards and Grants

- **Travel Grant:** Cognitive Science Arena, Brixen (Italy).
- **Travel Grant:** European Conference on Visual Perception, Mainz (Germany).
- **Travel Grant:** qHelp international seminar on Quantitative Psychology, Arrifana (Portugal).
- **Painter Award:** Best Poster, Cognitive Science Arena, Brixen (Italy).
- **Erasmus Placement Grant (2 Months):** Donders Sleep and Memory Lab, Nijmegen (The Netherlands).

- **Erasmus Placement Grant (5 Months):** Jaume I University, Castellón de la Plana (Spain).

Peer Reviews

Ad hoc reviewer (4 reviewed papers) for:

- Italian Journal of Applied Statistics (1)
- Frontiers in Psychology - Theoretical and Philosophical Psychology (1)
- Frontiers in Psychology - Cognition (1)
- PLoS ONE (1)

Teaching

Lectures on contract (docenze a contratto)

- **Inferenza e statistica Bayesiana** (10 hours). Master in Quantitative Psychology, Department of Philosophy, Sociology, Education and Applied Psychology, University of Padova. 21 July 2026 (planned).
- **Modelli lineari** (10 hours). Master in Quantitative Psychology, Department of Philosophy, Sociology, Education and Applied Psychology, University of Padova. 19 June 2026 (planned).
- **Test per le organizzazioni** (21 hours, 3 CFU). 2nd semester of the 2025/2026 academic year. Lecturer (titolare di modulo) for the Master's Degree in Psicologia Sociale, del Lavoro e della Comunicazione, University of Padova.
- **Introduction to meta-analysis with application in R** (5 hours, 1 CFU). Scheduled for the 2nd semester of the 2025/2026 academic year. Professor for the course within the PhD program in Psychological Sciences, University of Padova.

Seminars

[planned] means that the course has been officially scheduled but not delivered yet

- **Introduction to Bayesian Statistics with R and Stan** (30 hrs.). Intensive course on Bayesian statistics organized by the Italian Psychological Association (AIP). CEUB, Bertinoro (Italy), 29 June - 4 July 2026 (planned).
- **Introduction to R**. An intensive R programming course offered within the ARCA (Applied Research Course Academy) courses, Department of Developmental Psychology and Socialization, University of Padova. Material: <https://github.com/arca-dpss/course-R-2425>
 - December, 2024 (20 hrs.)
 - April, 2025 (20 hrs.)
 - November, 2025 (20 hrs.)

– March, 2026 (20 hrs.)

- **Learning in Dynamic Environments: A Tentative Implementation of the Volatile Kalman Filter in Stan.** Presentation (1 hr.) on a Bayesian-based implementation of a learning algorithm for the 2023/24 Psicostat meeting series: https://psicostat.dpss.psy.unipd.it/meetings/slides/2024-03-22_calderan.pdf

Integrative teaching (Didattica integrativa)

- **Assistance** during lectures for the course **Testing Psicologico** (Prof. Tatiana Marci), Bachelor's degree in Scienze psicologiche dello sviluppo, della personalità e delle relazioni interpersonali, University of Padova (14 hrs.).
- Preparation and correction of **practical group exercises** and exams for the course **Experimental Research Methods in Psychology** (Prof. Roberta Sellaro), Master's degree in Developmental and Educational Psychology, University of Padova (14 hrs.).
- **Assistance** in the preparation of exercises and corrections for the course **General Psychology** (Prof. Enrico Toffalini), Bachelor's degree in Techniques and Methods in Psychological Sciences, University of Padova (15 hrs.).
- **Assistance** during lectures and preparation of **practical group exercise** for the course **General Psychology** (Prof. Giovanna Mioni), Bachelor's degree in Social and Work Psychological Sciences, University of Padova (14 hrs.).

Workshops

- **Bayesian Approaches To Multiplicity.** Psicostat HandZone on Bayesian Approaches To Multiplicity, University of Padova, Padova (Italy). 27 December 2025 (1.5 hrs.). <https://psicostat.dpss.psy.unipd.it/pages/handzone.html>
- **Tempi di risposta.** Workshop on reaction time distributions, University of Padova, Padova (Italy). 10 June 2025 (1 hr.). Material: <https://mar-cald.github.io/tempi/>, <https://psicostat.dpss.psy.unipd.it/pages/handzone.html>

Academic Roles

- **Expert (Cultore della Materia) of Psychometrics (PSIC-01/C ex M-PSI/03)** Department of Developmental Psychology and Socialization, University of Padova, Italy. January 2026-present.
- **Representative of Research Fellows** in the Department of General Psychology, University of Padova. 02/2025 – 03/2026
- **Co-supervision of students** for their internships and Bachelor's (2) and Master's theses (1)

Organizational Activities

Methodological School: Replicability Crisis in Science

Member of the **Organizing team**. The school covers the epistemological, philosophical, and statistical foundations of replicability (<https://psicostat.github.io/replicability-psy-unipd-phd>).

- 6-11 June 2025, Padova (Italy)
- 18-20 May 2026, Padova (Italy)
- **Lecturers:** Prof. Branden Fitelson, Prof. Giovanni Parmigiani, Dr. Filippo Gambarota, Dr. Margherita Calderan (ed. 2026)

International Winter School - The 4 Ms: Meta-research, Measurement, Multi-lab, and Multiverse data analysis in Behavioral Sciences

Member of the **Organizing team**. The Winter School aims to introduce attendees to advancements in meta-research, providing both theoretical insights and practical tools to conduct rigorous research. 7-21 February 2025, Padova (Italy). (<https://winterschool4ms.dpg.unipd.it/>)

- **Lecturers:** Prof. Daniël Lakens, Prof. Dorothy Bishop, Prof. Jessica Flake, Prof. Lisa DeBruine, Prof. Ulrich Schimmack, Prof. Carlo Umiltà, Prof. Ettore Ambrosini, Dr. Filippo Gambarota, Dr. Giulia Calignano, Prof. Massimo Grassi, Prof. Livio Finos

ONIRICAMENTE

Member of the **Organizing team** of the one-day workshop dedicated to public outreach on the Psychology of Sleep. 21st June 2024, Padova (Italy) (<https://sites.google.com/view/oniricamente/home-page?authuser=0>)

Cognitive Neuroscience of Sleep Summer School

Member of the **Supporting team**. The Summer School examined methods for investigating sleep and memory in humans and animals. It also covered topics related to sleep medicine and dreaming. Nijmegen (Netherlands)

Third Mission (Terza Missione)

Replicability Crisis in Science – Methodological Seminars

University of Padova, Italy

2025

- The school aims to provide a broad and interdisciplinary perspective on replicability, along with tools that help participants explore aspects most relevant to their own field. Lectures cover the epistemological, philosophical, and statistical foundations of replicability and the so-called replication crisis (6-11/06/2025).

Replicability Crisis in Science – Methodological Seminars

University of Padova, Italy

2026

- The school aims to provide a broad and interdisciplinary perspective on replicability, along with tools that help participants explore aspects most relevant to their own field. Lectures cover the epistemological, philosophical, and statistical foundations of replicability and the so-called replication crisis (18-20/05/2026).

Science4all

University of Padova, Italy

2025

- Centra il bersaglio: tra misurazione, probabilità e incertezza. Activity organized by the Psicostat Research Group and proposed during the Science4All activity at the University of Padova (27-28/09/2025)

ONIRICAMENTE

University of Padova, Italy

2024

- One-day workshop dedicated to public outreach on the Psychology of Sleep (21/06/2024)

Research Groups Membership

Psicostat

University of Padova

2024 - Present

- I am a core team member of Psicostat, an interdisciplinary research group specializing in quantitative psychology, psychometrics, psychological testing, and statistics.
- <https://psicostat.dpss.psy.unipd.it/people.html>

SleePy

University of Padova

2019 - Present

- I am a member of SleepPy, a research group focusing on sleep psychology.

Collaborations

Prof. Jelle Goeman

Leiden University Medical Center, The Netherlands

2025 - Present

- Research on Bayesian approaches to the problem of multiplicity.

Prof. Aldo Solari

Ca' Foscari University of Venice, Italy

2025 - Present

- Research on Bayesian approaches to the problem of multiplicity.

Prof. Giovanni Parmigiani

Harvard University

2025 - Present

- Research and teaching projects about replicability in Psychology and Biomedical sciences.

Prof. Branden Fitelson

Northeastern University

2025 - Present

- Research and teaching projects about replicability in Psychology and Biomedical sciences.

Dr. Bryant Jongkees

Leiden University, The Netherlands

2024 - Present

- Research on stability and flexibility in cognitive control.

Dr. Antonino Visalli

University of Modena and Reggio Emilia, Italy

2023 - Present

- Research on computational models of learning and control.

Involvement in Research Groups

National Project on Malpractice in Psychological Research

Collaborative initiative aimed at identifying and addressing **methodological malpractice** in psychological science, with Enrico Toffalini (University of Padova), Filippo Gambarota (University of Padova) and Tommaso Feraco (University of Padova). My contributions include:

- Toffalini, E., **Calderan, M.**, Feraco, T., & Gambarota, F. Link Functions and Generalized Linear Models in Interaction Testing within Psychological Research. <https://osf.io/sjdu4>
- Toffalini, E., **Calderan, M.**, Pagan, R., Tobia, V., & Sonuga-Barke, E. Tails or Types? A Critical Systematic Review of Taxometric Studies of Neurodevelopmental Conditions. Under review in *JCPP Advances*. <https://github.com/EnricoToffalini/Taxometrics-Neurodevelopmental-Disorders/tree/main>

National Project on Adaptive Cognitive Control in Children

Part of a national research group focused on investigating adaptive cognitive control in typical, at-risk, and atypical populations within a Neuroconstructivist framework, with Giovanni Mento (University of Padova), Fiorella Del Popolo Cristaldi (University of Padova), Lisa Toffoli (University of Padova), Giulia Stefanelli (University of Padova), Gian Marco Duma (Epilepsy and Clinical Neurophysiology Unit, IRCCS E. Medea). The group has contributed to multiple peer-reviewed publications, conference presentations, and posters. My contributions specifically focused on the application of **Bayesian methods**, including **generalized multilevel models** and **survival analysis** techniques, in the following selected works:

- Toffoli, L., **Calderan, M.**, Del Popolo Cristaldi, F., Duma, G. M., Calcagni, A., Pastore, M., Tarantino, V., & Mento, G. (2025). Can I afford one more candy? How motivational contexts shape adaptive cognitive control in children. *Developmental Psychology*. Advance online publication. <https://doi.org/10.1037/dev0001976>.
- Stefanelli G., Toffoli L., **Calderan M.**, Del Popolo Cristaldi F., Duma G. M., Tarantino V., Visalli A., Sellaro R., Mento G. Apprendimento associativo implicito e volatilità ambientale nell'ADHD. **Video-Poster** at Giornate di aggiornamento sull'uso degli strumenti in Psicologia Clinica dello Sviluppo, Bologna (Italy), February 2025.
- Stefanelli G., Toffoli L., **Calderan M.**, Duma G. M., Del Popolo Cristaldi F., Tarantino V., Visalli A., Sellaro R. & Mento G. Super Learning Bros: Developmental trajectories of probabilistic learning in volatile environment. **Talk** at the Cognitive Science Arena, Brixen (Italy), February 2025.
- Toffoli, L., **Calderan, M.**, Duma, G. M., Frisina, S., Tarantino V., & Mento, G. I'm talking to you! Interactive digital content effects on children's adaptive cognitive control. **Talk** at the Cognitive Science Arena, Brixen (Italy), February 2024.

- Toffoli, L., **Calderan, M.**, Del Popolo Cristaldi, F., Duma, G. M., Calcagnì, A., Pastore, M., Tarantino, V., & Mento, G. Can I afford one more candy? How motivational contexts shape adaptive cognitive control in children. **Talk** at the Cognitive Science Arena, Brixen (Italy), February 2023.

Training

Model-based Neuroscience Summer School

University of Amsterdam, the Netherlands

29/07/2024 - 02/08/2024

- Hands-on experience in both cognitive modeling using evidence-accumulation models and cognitive neuroscience methods. Three-day introduction to evidence-accumulation modeling using Bayesian inference. Two days focused on how to use the evidence-accumulation framework to formalize performance in complex experimental paradigms, how to construct joint models of multiple tasks using techniques such as hierarchical factor models, and how to combine evidence-accumulation and reinforcement-learning models.
- Lecturers: Prof. Andrew Heathcote, Prof. Dora Matzke, Dr. Michael Nunez, Dr. Steven Miletic, and Dr. Niek Stevenson

qHelp international seminar on Quantitative Psychology

Arrifana, Portugal

26/03/2023 - 02/04/2023

- A one-week intensive seminar covering Shiny and R Markdown for interactive data reporting, Copula theory and its applications in dependence modeling, numerical methods, and optimization techniques for computational analysis.
- Lecturers: Julian Mollenhauer, Ana Lourenço, Prof. Luciano Stefanutti, Prof. Hans Colonius, Prof. Francis Tuerlinckx, and Prof. Sérgio Moreira

Statistical Rethinking

Online Course

06/01/2023 - 10/03/2023

- Pre-recorded lectures and weekly meetings to discuss the material. This course teaches data analysis, focusing on scientific models: conceptual and causal models. Bayesian data analysis to connect scientific models to evidence.
- Lecturer: Prof. Richard McElreath

Fundamentals and Applications of Pupillometry

University of Padova, Italy

20 hours

- This course offers an introduction to the use of pupillometry in cognitive science, combining practical exercises in data processing with free software tools: OpenSesame for implementing

classic cognitive psychology tasks and R for analyzing and visualizing continuous pupil diameter changes. Covering key topics such as noise sources, experimental design, data processing options, and the cognitive interpretation of pupillometric measures.

- Professor: Dr. Giulia Calignano

EEGLAB: analysis of electroencephalographic data

University of Padova, Italy

20 hours

- The EEGLAB and ERPLAB toolboxes, which operate within MATLAB, facilitate the analysis of electroencephalographic (EEG) data. This course offers foundational skills in data analysis and theoretical knowledge through practical examples, covering key topics such as data pre-processing and event-related potential (ERP) computation.
- Professor: Dr. Mattia Doro

Introduction to R

University of Padova, Italy

20 hours

- This course introduces the fundamentals of the open-source programming language R, teaching core programming concepts, data structures, and function creation through practical exercises to build versatile skills applicable across academic and professional settings.
- Professor: Dr. Filippo Gambarota

PsychoPy3: experimental programming

University of Padova, Italy

20 hours

- This course introduces participants to PsychoPy, an open-source Python-based software for designing and running behavioral science experiments, teaching them how to create, test, deploy, and manage experiments using its flexible and user-friendly graphical interface.
- Professor: Dr. Thomas Quettier

Scientific Societies Membership

Cognitive Science Society

2024 - Present

Italian Reproducibility Network

2023 - Present

European Sleep Research Society

2023 - Present

AIP, Sezione Sperimentale

2022 - Present

Society for the Improvement of Psychological Science

2022 - 2023

European Society for Cognitive Psychology

2021 - 2023

Languages

Italian

Mother Tongue

English

Proficient

Spanish

Beginner

Programming Languages

R / RStudio

Proficient

Stan

Intermediate

JAGS

Basic

Python

Basic

MATLAB

Basic

Research and Project Management Skills

PsychoPy/Pavlovia and OpenSesame/OSWeb

Intermediate

- Task development for behavioral research.

GitHub and GitLab

Intermediate

- Version control and collaboration.
- <https://github.com/Mar-Cald>, <https://gitlab.pavlovia.org/mar.cald>

Docker

Basic

- Development and deployment of containers.

Google Suite, Office Suite

Intermediate

- Gmail, Calendar, Drive, Meet, Docs, Forms, Sheets.

R Packages

- *multibayes*: tools for Bayesian multiplicity adjustments (**maintainer**). <https://mar-cald.github.io/multibayes/>

Operating Systems

macOS

Intermediate

Windows

Intermediate

Linux

Intermediate

I authorize the processing of personal data included in my CV in accordance with Legislative Decree 101/2018 and the GDPR (Regulation EU 2016/679). I hereby declare that the information and details provided in this Curriculum Vitae are true and accurate to the best of my knowledge, acknowledging the legal and criminal consequences of providing false statements as per Article 76 of Presidential Decree 445/2000.

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